



Department of Computer Science and Engineering
Green University of Bangladesh
Midterm Examination, Fall 2024
CSE 301, Web Programming



Duration: 1.5 hours

Total Marks: 30

Answer all of the following questions.

(The CO# along with each sub-question represents the mapping of the questions onto the expected outcomes of the course provided in the course outline.)

Question 1.

a. CO1 3 Marks
"Inline element can contain block elements" - evaluate this statement as either true or false.

— or —

Analyze the attributes that will allow the developer to change the font-related items and add custom gaps between the lines.

b. CO1 4 Marks
Illustrate two files index.html and main.css to develop the website given in Figure-1.

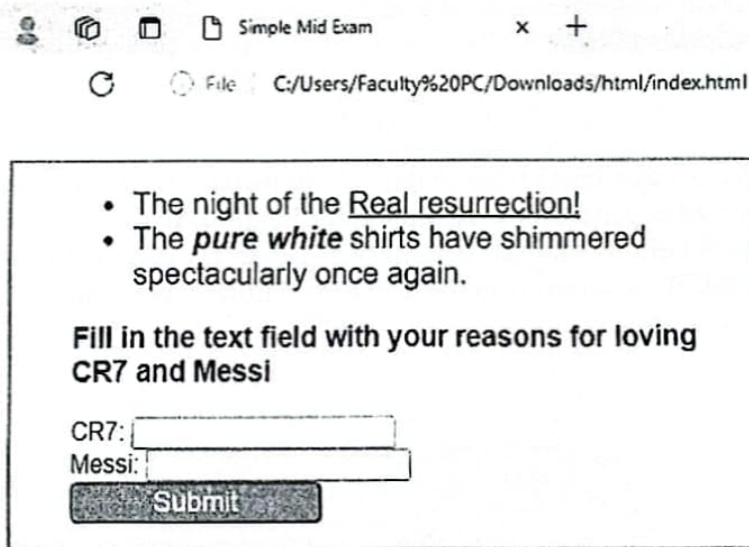


Figure 1: A Simple HTML Page

c. CO1 3 Marks
Imagine you're designing a webpage and need to apply specific styles to certain elements. You want to apply style to a unique component, such as a header, and also apply the same style to multiple sections on the page. In this scenario, differentiate between using an ID and a class to achieve your styling goals. Represent with examples.

Question 2.

a.

2 Marks

Interpret the components of the CSS declaration syntax written as following by identifying the actual names of X, Y, and Z. Discuss the proper way of writing CSS syntax with an example.

```

1  X {
2      Y: Z;
3  }
```

— or —

Explain the differences between null and undefined in JavaScript with a proper example.

b.

3 Marks

Discuss the ways to minimize the risks associated with global variables.

c.

2 Marks

Demonstrate the output of the following code snippet:

```

1  function greet() {
2      console.log("Hello!");
3      console.log("Null");
4  }
5  let greeting = greet();
6  console.log(greeting);
```

d.

3 Marks

Describe the role of anonymous functions in JavaScript and identify the benefits and drawbacks when used in event-driven programming.

Question 3.

a.

4 Marks

The .box element will have a dynamic width that is 50% of its parent (body in this case) but will not shrink below 300px or grow beyond 600px. Although the height is fixed at 200px but has constraints. The lowest and maximum heights that can be observed are 150px and 400px, respectively. Demonstrate a relatable external CSS depending on the provided Figure-2, keeping the mentioned scenarios in mind.

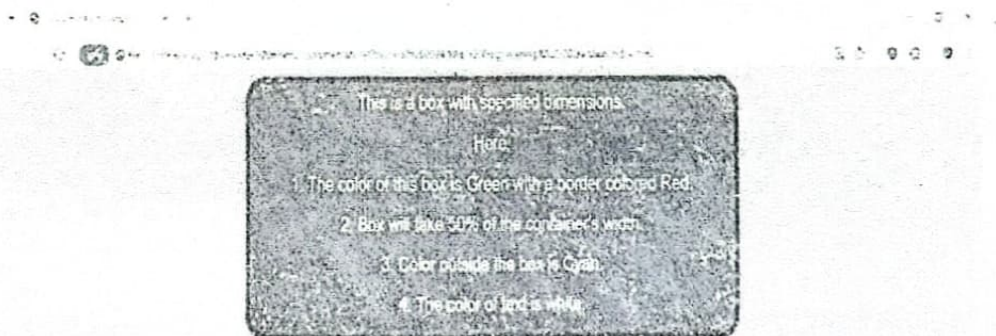


Figure 2: A Simple Web Page

— or —

Different types of browser can be used to access any web application. Like Microsoft Explorer, some browsers do not support all the updated functionality. For this reason, developers need to write Browser-specific code. Now, identify the pros and cons of this approach and analyze the effect in the compatibility and future-proofing of the web applications.

b**CO2 4 Marks**

```
1 function authenticateUser(inputUsername, inputPassword) {  
2     var storedUsername = "admin";  
3     var storedPassword = 12345;  
4     if (inputUsername == storedUsername && inputPassword ==  
5         storedPassword) {  
6         return "Authentication successful!";  
7     } else if (inputUsername === storedUsername && inputPassword ===  
8         storedPassword) {  
9         return "Authentication successful with strict check!";  
10    } else {  
11        return "Authentication failed!";  
12    }  
13 }  
14 console.log(authenticateUser("admin", "12345")); // Case 1  
15 console.log(authenticateUser("admin", 12345)); // Case 2  
16 console.log(authenticateUser("Admin", 12345)); // Case 3  
17 console.log(authenticateUser("admin", 123)); // Case 4
```

c**CO2 2 Marks**

Revise the function (authenticateUser) of Question 3(b) so that authentication only succeeds when both the value and type match exactly.

□ End of Question □



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Question 1.



CO1 2 Marks

Find the errors from the following code segment and write the corrected parts.

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4
5 </head>
6 <body>
7   <div>In academic publishing, a paper is an academic work that is
   usually published in an academic journal.
8   <section>In academic publishing, a paper is an academic work that
   is usually published in an
9   <h1>Academic Journal.</h1>
10  <p>The most common formats of transmission are <em> <i>HTML, PDF
   </em></i> and, </div> in specialized fields like mathematics
   and physics, TeX and </p>PostScript.</div>
11 </body>
12 </html>
```



CO1 3 Marks

Imagine you are creating an online shopping website. Discuss the HTML list structures you would use to present categories of products, including subcategories and product details such as prices and descriptions.

— OR —

"The box model is only used for content representation in CSS." Explain your answer with the help of the border, margin, and padding properties.



CO1 3 Marks

Apply basic HTML and CSS for the following TABLE.

Time Table					
Hours	Mon	Tue	Wed	Thu	Fri
	Science		Math	Science	Arts
	Lunch				
	Math	Science	Math	Project	
	Math	Science	Math		

```
28 <p>They are usually described as early, second early, and  
maincrop potatoes.</p>  
29 </body>  
30 </html>
```

Q.2

CO1 3 Marks

Calculate the total width of the box using the knowledge of the CSS Box Model. Explain the output briefly.

```
1 <!DOCTYPE html>  
2 <html>  
3 <head>  
4 <style>  
5   .box {  
6     width: 100px;  
7     height: 50px;  
8     padding: 5px;  
9     border: 1px solid red;  
10    background color: red;  
11  }  
12  
13 </style>  
14 </head>  
15 <body>  
16   <div class="box">  
17     <p>Paragraph  
18       <span>This is text with</span>  
19       lines  
20       <span>top alignment.</span>  
21     </p>  
22   </div>  
23  
24  
25 </body>  
26 </html>
```

Question 3.

Q.3

CO2 5 Marks

Write a JavaScript function named `updateUI` that executes a given callback function every `x` milliseconds, where `x` is a parameter to `updateUI`. The callback function, named `fetchAndUpdate`, simulates fetching data from a server and updating the UI. Assume `fetchAndUpdate` is already implemented and simply needs to be called. Your function should also ensure that it can be stopped/cleared when needed.

— OR —

Suppose, You are developing a feature for a client's website that aims to make the site more interactive and personal to its visitors. The feature should greet users by their name if they provide it; otherwise, it should fall back to a generic greeting. Now You have to implement a prototype using basic HTML and JavaScript, to develop the feature provided by the scenario.